

AI Unit 1

4 Marks Questions

1. Define Artificial Intelligence.

1. Artificial Intelligence (AI) is a branch of computer science that focuses on creating intelligent machines. These machines are designed to perform tasks that normally require human intelligence such as learning, reasoning, and decision-making.
2. AI systems are capable of perceiving their environment and analyzing data. They use this information to make decisions that help achieve specific goals.
3. According to John McCarthy, AI is “the science and engineering of making intelligent machines.” It mainly deals with developing intelligent computer programs.
4. The main objective of AI is to simulate human intelligence in machines. It allows systems to function autonomously and adapt to new situations.

2. List any four applications of AI.

1. In healthcare, AI is used for disease diagnosis and medical imaging. It helps doctors detect diseases early and improves the accuracy of treatment.
2. In education, AI supports intelligent tutoring systems and personalized learning. It helps students learn at their own pace and provides customized feedback.
3. In finance, AI is used for fraud detection and algorithmic trading. It analyzes large amounts of data to identify suspicious transactions and predict market trends.
4. In transportation, AI is used in autonomous vehicles and traffic management systems. It improves road safety and reduces traffic congestion.

3. What are the components of an AI program?

1. Learning is a major component of AI. It allows machines to improve their performance by analyzing data and gaining experience.
2. Reasoning helps AI systems make logical decisions. It enables them to draw conclusions from available information.
3. Problem-solving allows AI to identify solutions for complex tasks. It uses search algorithms and optimization techniques to find the best result.
4. Perception enables AI to understand its surroundings. It processes inputs such as images, speech, and other sensory data.
5. Language processing helps machines understand and generate human language. It is used in chatbots and virtual assistants.

4. Define an Intelligent System.

1. An Intelligent System is a system that can perceive its environment and interpret information. It uses this information to make decisions and take appropriate actions.
2. It has the ability to learn from experience and improve over time. This helps the system adapt to new situations without constant human control.
3. Intelligent systems can reason and solve problems efficiently. They analyze data and select the best possible action.
4. These systems operate autonomously to achieve specific goals. Examples include self-driving cars and expert systems.

5. List any four sub-areas of AI.

1. Machine Learning (ML) is a sub-area of AI that enables systems to learn from data. It improves performance without being explicitly programmed.
2. Natural Language Processing (NLP) helps computers understand and generate human language. It is used in chatbots and language translation systems.
3. Computer Vision allows AI to interpret images and videos. It is used in facial recognition and autonomous vehicles.
4. Robotics combines AI with mechanical systems to create intelligent machines. Robots can sense, act, and learn from their environment.

6. Write short note on the History of Artificial Intelligence.

1. The history of Artificial Intelligence began in the 20th century with early ideas about intelligent machines. In 1950, Alan Turing introduced the Turing Test and asked the question, "Can machines think?"
2. In 1956, John McCarthy officially coined the term "Artificial Intelligence." This year is considered the formal birth of AI as a research field.
3. During the 1960s and 1970s, important developments took place such as ELIZA, the first chatbot, and Shakey, the first intelligent robot. These systems showed that machines could simulate conversation and perform reasoning tasks.
4. In 1997, IBM's Deep Blue defeated world chess champion Garry Kasparov. This milestone proved that AI could outperform humans in complex strategic tasks.

7. Define Natural Language Processing (NLP).

1. Natural Language Processing (NLP) is a sub-area of Artificial Intelligence that enables computers to understand, interpret, and generate human language. It helps machines communicate with humans in a natural way.
2. NLP involves tasks such as syntax analysis and semantic analysis. These processes help machines understand sentence structure and meaning.
3. It is widely used in applications like chatbots, virtual assistants, and language translation systems. NLP allows systems to respond correctly in conversations.
4. By combining linguistics and machine learning, NLP improves interaction between humans and computers. It plays an important role in modern AI systems.

8. Mention any four current trends in AI.

1. Intelligent Process Automation (IPA) is a major trend in AI. It combines Robotic Process Automation and AI to automate business processes efficiently.
2. AI-powered cybersecurity is becoming increasingly important. It helps detect cyber threats and prevent data breaches using advanced algorithms.
3. Autonomous vehicles are a growing trend in AI. Companies are investing heavily in self-driving technology to improve road safety and convenience.
4. Explainable AI is gaining attention due to trust issues in machine learning models. It aims to make AI decisions more transparent and understandable to users.

9. What is Expert System?

1. An Expert System is a type of AI system that mimics the decision-making ability of a human expert. It uses knowledge and rules to solve specific problems in a particular domain.
2. It consists of a knowledge base and an inference engine. The knowledge base stores facts and rules, while the inference engine applies logical reasoning to reach conclusions.
3. Expert systems are used in areas such as medical diagnosis, financial advising, and troubleshooting. They help in making accurate and consistent decisions.
4. These systems can explain their reasoning to users. This makes them useful for knowledge-intensive tasks.

10. List any four characteristics of an intelligent system.

1. An intelligent system has the ability to learn from experience. It improves its performance over time by analyzing data and adapting to changes.
2. It has reasoning and problem-solving capabilities. The system can analyze situations and make logical decisions.
3. It can perceive its environment using sensory inputs such as images and sounds. This helps it interact effectively with the real world.
4. An intelligent system can act autonomously to achieve specific goals. It operates without constant human supervision and adapts to new situations.

5 Marks Questions

1. Explain the history and evolution of Artificial Intelligence.

1. The history of Artificial Intelligence began in the early 20th century with the idea of intelligent machines. In 1950, Alan Turing introduced the Turing Test and raised the important question, “Can machines think?”
2. In 1956, John McCarthy coined the term “Artificial Intelligence” at the Dartmouth Conference. This year is considered the official birth of AI as a research field.
3. During the 1960s and 1970s, systems like ELIZA and Shakey were developed. ELIZA demonstrated natural language conversation, while Shakey became the first intelligent robot capable of reasoning and movement.
4. In 1997, IBM’s Deep Blue defeated world chess champion Garry Kasparov. This achievement proved that AI systems could outperform humans in complex strategic tasks.
5. In the 2000s and beyond, AI entered daily life through virtual assistants, machine learning, autonomous vehicles, and robotics. Today, AI continues to evolve rapidly with advancements in deep learning and automation.

2. Describe the foundations of AI.

1. Mathematics forms the core foundation of AI. Concepts like linear algebra, probability, statistics, and optimization help in building and improving AI models.
2. Computer Science provides algorithms, data structures, and programming techniques. It helps in designing efficient systems and solving computational problems.
3. Cognitive Science and Psychology contribute by studying human learning and decision-making. AI systems are designed by understanding how humans think and solve problems.

4. Linguistics plays an important role in Natural Language Processing. It helps machines understand grammar, sentence structure, and meaning.
5. Philosophy and Ethics guide AI in logical reasoning and responsible decision-making. They ensure fairness, transparency, and proper knowledge representation in intelligent systems.
6. Neuroscience inspires neural networks by studying the human brain. It helps in developing learning models similar to biological systems.

3. Explain the components of an AI program.

1. Learning is a key component of AI systems. It enables machines to improve performance over time using data and experience.
2. Reasoning allows AI to make logical decisions. It uses deductive, inductive, or abductive reasoning to draw conclusions.
3. Problem-solving helps AI find solutions to complex challenges. It uses search algorithms and optimization techniques to identify the best possible outcome.
4. Perception allows AI to understand sensory inputs such as images and speech. Technologies like computer vision and speech recognition are part of perception.
5. Language processing enables machines to understand and generate human language. It is used in chatbots, translation systems, and virtual assistants.
6. Data acts as the fuel for AI systems. The quality and quantity of data directly affect the accuracy and fairness of AI models.

4. Discuss different sub-areas of AI.

1. Machine Learning (ML) is a major sub-area of AI. It enables systems to learn from data without explicit programming.
2. Natural Language Processing (NLP) focuses on understanding and generating human language. It is used in chatbots, voice assistants, and translation systems.
3. Computer Vision allows machines to interpret images and videos. It is used in facial recognition, surveillance, and autonomous vehicles.
4. Robotics combines AI with mechanical systems to design intelligent robots. These robots can sense, act, and adapt to their environment.
5. Expert Systems use knowledge bases and inference engines to mimic human expertise. They are applied in medical diagnosis, financial analysis, and troubleshooting.

5. Explain the categorization of intelligent systems with suitable examples.

1. Intelligent systems are categorized based on capabilities into Narrow AI, General AI, and Super AI. Narrow AI is designed for specific tasks such as Siri or recommendation systems.
2. General AI refers to systems that can perform any intellectual task like a human. Currently, such systems do not exist and remain under research.
3. Super AI is a hypothetical concept where machines surpass human intelligence. It would exceed human reasoning and decision-making abilities.
4. Intelligent systems are also categorized based on functionality into Reactive Machines, Limited Memory, Theory of Mind, and Self-Awareness.
5. Reactive machines like IBM's Deep Blue respond only to current inputs without memory. Limited memory systems such as self-driving cars use past data temporarily to make decisions.

6. Describe various applications of AI in real-world scenarios.

1. In healthcare, AI is used for disease diagnosis and medical imaging. It helps doctors detect diseases early and improves treatment accuracy.
2. In education, AI supports intelligent tutoring systems and personalized learning platforms. These systems adapt content based on student performance.
3. In finance, AI is applied in fraud detection and algorithmic trading. It analyzes large volumes of financial data to detect suspicious patterns.
4. In transportation, AI is used in autonomous vehicles and traffic management systems. It improves road safety and reduces congestion.
5. In e-commerce, AI powers recommendation systems and customer support chatbots. These systems enhance user experience and increase sales.
6. In security and surveillance, AI enables facial recognition and threat detection. It strengthens safety measures in public and private sectors.

7. Explain the characteristics of an Intelligent System.

1. An intelligent system has the ability to perceive its environment. It collects data through sensors such as cameras or microphones and interprets the information.
2. It can learn from experience and improve over time. Learning allows the system to adapt to new situations without being explicitly programmed.
3. An intelligent system possesses reasoning and decision-making abilities. It analyzes available information and selects the most appropriate action.
4. It can solve complex problems using search and optimization techniques. This enables it to handle real-world challenges effectively.
5. It operates autonomously to achieve specific goals. The system functions with minimal human intervention.
6. Intelligent systems can interact with humans using natural language. This makes communication smooth and user-friendly.

8. Discuss current trends in AI such as Deep Learning and Generative AI.

1. Deep Learning is a major trend in AI that uses multi-layered neural networks. It can automatically extract features from raw data and handle complex tasks like image and speech recognition.
2. Generative AI is another emerging trend that creates new content such as text, images, and music. It is used in chatbots, content generation, and creative applications.
3. Intelligent Process Automation combines AI with automation tools. It improves efficiency in business operations by handling repetitive tasks.
4. AI-powered cybersecurity is gaining importance due to rising cyber threats. It helps detect and prevent attacks using pattern recognition and predictive analysis.
5. Explainable AI is becoming essential to ensure transparency. It makes AI decisions more understandable and trustworthy.

9. Differentiate between Artificial Intelligence and Machine Learning.

1. Artificial Intelligence is a broad field that focuses on creating intelligent machines. It includes various techniques that enable machines to mimic human intelligence.
2. Machine Learning is a sub-area of AI. It allows systems to learn from data and improve performance without explicit programming.
3. AI covers reasoning, perception, problem-solving, and language processing. Machine Learning mainly focuses on data-driven learning algorithms.
4. AI can include rule-based systems like expert systems. Machine Learning relies on training models using data sets.
5. In simple terms, AI is the overall concept of intelligent systems, while Machine Learning is one method used to achieve AI.

10. Explain how AI is used in healthcare and finance sectors.

1. In healthcare, AI is used for medical imaging and disease detection. It helps identify conditions such as cancer and heart disease with high accuracy.
2. AI supports drug discovery and research. It analyzes biological data to speed up the development of medicines.
3. In hospital management, AI analyzes patient records and resource utilization. This improves efficiency and patient care.
4. In finance, AI is used for fraud detection. It monitors transactions in real time and detects unusual activities.
5. AI also helps in algorithmic trading and risk analysis. It studies market trends and supports better investment decisions.

10 Marks Questions

1. Explain the concept of Intelligent Systems. Discuss the categorization of intelligent systems with examples.

1. An Intelligent System is a system that can perceive its environment, interpret information, learn from experience, and act autonomously to achieve specific goals. It mimics human intelligence by performing tasks such as reasoning, decision-making, and problem-solving.
2. Intelligent systems are categorized based on capabilities into Narrow AI, General AI, and Super AI. Narrow AI is designed to perform specific tasks such as virtual assistants, recommendation systems, or chess-playing programs.
3. General AI refers to systems that can perform any intellectual task similar to a human. Currently, General AI does not exist and remains a research goal for scientists worldwide.
4. Super AI is a hypothetical stage where machines surpass human intelligence in all cognitive aspects. It is considered a future concept and has not yet been developed.
5. Intelligent systems are also categorized based on functionality into Reactive Machines, Limited Memory, Theory of Mind, and Self-Awareness. Reactive machines like Deep Blue respond only to current inputs, while Limited Memory systems such as self-driving cars use past data temporarily for decision-making.
6. Theory of Mind and Self-Awareness AI are advanced theoretical categories. These systems would understand emotions or possess consciousness, but they do not yet exist.

2. Describe the components of an AI program. Explain how these components work together in developing an AI system.

1. Learning is a core component of an AI program. It enables systems to improve performance by analyzing data and identifying patterns through methods such as supervised or reinforcement learning.
2. Reasoning allows AI to make logical decisions based on available information. It uses deductive, inductive, or abductive reasoning to draw meaningful conclusions.
3. Problem-solving helps AI find optimal solutions to complex challenges. It uses search algorithms and optimization techniques to select the best possible action.
4. Perception enables AI systems to understand sensory inputs like images, speech, or text. Technologies such as computer vision and speech recognition are part of perception.
5. Language processing allows machines to understand and generate human language. It plays an important role in chatbots and virtual assistants.
6. Data acts as the fuel for all AI components. High-quality data ensures accuracy, fairness, and effective decision-making.
7. All these components work together to build a complete AI system. Perception gathers input, learning improves performance, reasoning analyzes information, and problem-solving selects actions to achieve desired goals.

3. Discuss the major sub-areas of Artificial Intelligence.

1. **Machine Learning (ML)** is one of the most important sub-areas of AI. It enables systems to learn from data and improve their performance without being explicitly programmed. ML includes supervised, unsupervised, and reinforcement learning techniques. It is widely used in spam detection, recommendation systems, and predictive analytics.
2. **Natural Language Processing (NLP)** allows computers to understand, interpret, and generate human language. It involves syntax analysis, semantic analysis, and dialogue management. NLP is used in chatbots, virtual assistants, sentiment analysis, and language translation applications.
3. **Robotics** combines AI with mechanical systems to design intelligent machines. Robots can sense their environment, process information, and perform tasks autonomously. Robotics is used in manufacturing, medical surgery, space exploration, and delivery systems.
4. **Computer Vision** enables machines to interpret images and videos. It performs tasks like object detection, facial recognition, and image classification. Applications include surveillance systems, self-driving cars, and medical imaging.
5. **Expert Systems** are AI systems that mimic human experts using a knowledge base and inference engine. They provide decisions or recommendations in specific domains like medical diagnosis, financial advising, and troubleshooting.

4. Explain the applications of Artificial Intelligence in various domains with examples.

1. In **Healthcare**, AI is used for disease diagnosis, medical imaging, and drug discovery. It helps doctors detect diseases like cancer at early stages and improves treatment accuracy.
2. In **Education**, AI supports intelligent tutoring systems and personalized learning platforms. These systems analyze student performance and adapt teaching methods accordingly.
3. In **Finance**, AI is applied in fraud detection, risk analysis, and algorithmic trading. It monitors transactions in real time and predicts market trends based on historical data.
4. In **Transportation**, AI powers autonomous vehicles and traffic management systems. It improves road safety, reduces accidents, and optimizes traffic flow.
5. In **Manufacturing**, AI is used for predictive maintenance and robotics. It helps detect equipment faults early and increases production efficiency.
6. In **E-commerce**, AI enables recommendation systems and customer support chatbots. It enhances user experience by suggesting products based on customer preferences.
7. In **Security**, AI is used for facial recognition and surveillance systems. It strengthens public safety and prevents unauthorized access.

5. Discuss the current trends in AI and their impact on industry and society.

1. **Deep Learning** is one of the most significant trends in AI. It uses multi-layered neural networks to process complex data such as images, audio, and text. This has improved applications like speech recognition, medical diagnosis, and image classification with high accuracy.
2. **Generative AI** is rapidly growing in popularity. It can generate new content such as text, images, music, and even software code. This trend is transforming industries like media, education, entertainment, and marketing by increasing creativity and productivity.
3. **Intelligent Process Automation (IPA)** combines AI with robotic process automation. It helps organizations automate repetitive and rule-based tasks, improving efficiency and reducing operational costs.
4. **AI-powered Cybersecurity** is becoming increasingly important. AI systems detect unusual patterns and prevent cyberattacks, protecting sensitive data and improving digital safety.
5. **Autonomous Vehicles** are a major technological trend. Self-driving technology aims to reduce accidents and improve transportation systems.
6. These trends significantly impact society by improving productivity, innovation, and service quality. However, they also raise concerns related to job displacement, privacy, bias, and ethical decision-making.

6. Explain the foundations of AI in detail.

1. **Mathematics** is the backbone of Artificial Intelligence. Concepts like linear algebra, probability, statistics, and optimization are used to design models and handle uncertainty in decision-making.
2. **Computer Science** provides algorithms, programming techniques, and data structures necessary for implementing AI systems. It ensures efficient data processing and problem-solving capabilities.
3. **Psychology and Cognitive Science** contribute by studying how humans learn, think, and make decisions. AI systems are inspired by human learning models and reasoning patterns.
4. **Linguistics** plays a crucial role in Natural Language Processing. It helps AI understand grammar, syntax, semantics, and contextual meaning of language.
5. **Philosophy** contributes logic, reasoning principles, and ethical guidelines. It addresses questions about intelligence, knowledge representation, fairness, and responsible AI design.
6. These disciplines work together to form the foundation of AI. Artificial Intelligence is therefore an interdisciplinary field combining science, mathematics, logic, and human understanding

7. Identify the sub-area of AI used in:

- **Chatbots**

Chatbots mainly use **Natural Language Processing (NLP)**. NLP enables computers to understand, interpret, and generate human language for effective communication.

It includes tasks such as syntax analysis, semantic analysis, and dialogue management. Through NLP, chatbots can respond to user queries, answer questions, and simulate human-like conversations. Some advanced chatbots also use Machine Learning to improve responses over time.

- **Self-driving cars**

Self-driving cars primarily use **Computer Vision** and **Machine Learning**. Computer Vision helps the system interpret images and videos from cameras to detect objects, traffic signals, pedestrians, and road signs. Machine Learning enables the vehicle to learn from driving data and make accurate decisions. These systems often use Limited Memory AI to temporarily store recent information such as speed and distance.

- **Face recognition systems**

Face recognition systems mainly use **Computer Vision**. This sub-area allows AI to detect, analyze, and identify human faces from images or video inputs. Machine Learning algorithms are also used to match facial patterns with stored data. It is widely applied in security, surveillance, and biometric authentication systems.